Fourier Analysis and Regulated DC Power Supply

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***Abstract*—This document introduces periodic signals, Fourier series, and Fourier transforms, and applies these concepts to filtering. It also presents the design considerations and measured results for a regulated DC power-supply and USB output circuit.**

1. Periodic Function

1.1. Consider the full-wave-rectified sinusoidal signal

$$x(t) = A_0 \left|\sin\left(2\pi f_0 t\right)\right|, \tag{1.1}$$

where A_0 is the peak amplitude and f_0 is the frequency of the underlying sinusoid. The absolute value makes the signal repeat twice per cycle of the underlying sinusoid.

1.1 Plot $x(t)$.

1.2 Show that $x(t)$ is periodic and find its period. In particular, distinguish the period of $x(t)$ from the period of the underlying sinusoid.

2. Fourier Series

Consider $A_0 = 12\,\text{V}$ and $f_0 = 50\,\text{Hz}$ for all numerical calculations. The rectified signal has fundamental frequency $2f_0 = 100\,\text{Hz}$.

2.1 If

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j2\pi k f_0 t} \tag{2.1}$$

show that

$$c_k = f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) e^{-j2\pi k f_0 t}\, dt \tag{2.2}$$

2.2 Find c_k for (1.1). In particular, account for the fact that the signal has period $1/(2f_0)$, although the complex series is written using the underlying frequency f_0.

2.3 Verify (1.1) using Python.

2.4 Show that

$$x(t) = \frac{a_0}{2} + \sum_{k=1}^{\infty} \left(a_k \cos(2\pi k f_0 t) + b_k \sin(2\pi k f_0 t)\right) \tag{2.3}$$

and obtain the coefficients

$$a_0 = 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t)\, dt,$$

$$a_k = 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) \cos(2\pi k f_0 t)\, dt,$$

$$b_k = 2f_0 \int_{-\frac{1}{2f_0}}^{\frac{1}{2f_0}} x(t) \sin(2\pi k f_0 t)\, dt, \quad k$$

$geq1$.

2.5 Find a_k and b_k for (1.1). Use the same coefficient convention as in the real Fourier series above.

2.6 Verify (2.3) using Python.

For this signal, the coefficients provide a useful consistency check: $b_k = 0$ for every k, $a_0/2 = 2A_0/\pi$, and $a_{2n} = -4A_0/[\pi(4n^2-1)]$ for $n \geq 1$, while $a_k = 0$ for odd k. Equivalently, $c_0 = 2A_0/\pi$, $c_{2n} = c_{-2n} = -2A_0/[\pi(4n^2-1)]$, and all odd-indexed complex coefficients vanish.

3. Fourier Transform

3.1 The Dirac delta is a generalized function characterized by

$$\delta(t) = 0, \quad t \neq 0, \tag{3.1}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1, \tag{3.2}$$

$$\int_{-\infty}^{\infty} g(t)\delta(t - t_0)\, dt = g(t_0). \tag{3.3}$$

3.2 The Fourier Transform of $g(t)$ is

$$G(f) = \int_{-\infty}^{\infty} g(t) e^{-j2\pi f t}\, dt. \tag{3.4}$$

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3.3 Show that

$$g(t - t_0) \overset{\mathcal{F}}{\longleftrightarrow} G(f)e^{-j2\pi f t_0}. \tag{3.5}$$

3.4 Show that

$$G(t) \overset{\mathcal{F}}{\longleftrightarrow} g(-f). \tag{3.6}$$

3.5 Determine the Fourier transform of $\delta(t)$, treating it as a generalized function (distribution).
3.6 Determine the Fourier transform of $e^{-j2\pi f_0 t}$.
3.7 Determine the Fourier transform of $\cos(2\pi f_0 t)$.
3.8 Find the Fourier transform of the periodic signal $x(t)$ and plot its line spectrum. Use the distributional Fourier-transform convention and verify the result using Python.
3.9 Using the normalized definitions $\mathrm{rect}\,(t) = 1$ for $|t| < 1/2$ and $\mathrm{sinc}\,(t) = \sin(\pi t)/(\pi t)$, show that

$$\mathrm{rect}\,(t) \overset{\mathcal{F}}{\longleftrightarrow} \mathrm{sinc}\,(f)\,. \tag{3.7}$$

Verify using Python.
3.10 Determine the Fourier transform of $\mathrm{sinc}\,(t)$ using the same normalized convention. Verify using Python.

4. Filter

4.1 Determine the required DC gain and the harmonic-rejection condition on $H(f)$ so that the steady-state output is a constant 5 V. State any assumptions about the filter passband.
4.2 For an ideal low-pass model with cutoff f_c satisfying $0 < f_c < 100\,\mathrm{Hz}$, find the corresponding impulse response $h(t)$.
4.3 Verify your result using convolution.

The average value of the input is $X_{\mathrm{DC}} = 2A_0/\pi$. For $A_0 = 12\,\mathrm{V}$, the required DC gain is therefore $K = 5/(24/\pi) = 5\pi/24 \approx 0.6545$. An ideal low-pass response with $H(f) = K$ for $|f| < f_c$ and $H(f) = 0$ for $|f| > f_c$, where $0 < f_c < 100\,\mathrm{Hz}$, passes the DC component and rejects the first and all higher harmonics. With the normalized sinc definition, its impulse response is $h(t) = 2Kf_c\mathrm{sinc}\,((2f_c t))$.

5. Filter Design

5.1 Design a Butterworth approximation to the specified low-pass response $H(f)$. State the filter order, cutoff frequency, and any design tolerance used.
5.2 Design a Chebyshev approximation to the specified low-pass response $H(f)$. State the filter order, passband ripple, and cutoff frequency.
5.3 Design a circuit for your Butterworth filter.
5.4 Design a circuit for your Chebyshev filter.

6. Circuit Input and Transformer

The circuit considered here uses a nominal primary supply of 220 V RMS at 50 Hz and a measured secondary voltage of 10.53 V RMS. The transformer reduces the AC voltage and provides galvanic isolation.

TABLE I: Input and transformer values

Quantity	Value
Primary RMS voltage	220 V
Mains frequency	50 Hz
Primary peak voltage (calculated)	311.1 V
Measured secondary RMS voltage	10.53 V
Secondary peak voltage (calculated)	14.89 V
Approximate turns ratio	20.9 : 1

For a sinusoidal waveform, the peak voltage is $\sqrt{2}$ times the RMS voltage. The ideal transformer voltage ratio equals its turns ratio. These values define the transformer operating point used in the calculations below.

A separate centre-tapped transformer configuration is also considered. It uses a 12–0–12 V centre-tapped transformer, a 12 V RMS winding output, and an approximately 18 V peak level. This is a separate transformer configuration, not the 10.53 V RMS secondary used for the calculations above.

7. Full-Wave Bridge Rectifier

The bridge rectifier uses four diodes. Two diodes conduct during each half-cycle, so the current through the load remains in the same direction. The rectified waveform has twice the input frequency.

For a first-order bridge-rectifier calculation, the forward drop of each conducting silicon diode is taken as 0.7 V. Since two diodes conduct in each half-cycle, the approximate output peak is $14.89 - 2(0.7) = 13.49\,\mathrm{V}$. Approximating the resulting waveform as a full-wave-rectified sinusoid with this peak gives an average value of $2(13.49)/\pi = 8.59\,\mathrm{V}$ and an RMS value of $13.49/\sqrt{2} = 9.54\,\mathrm{V}$. These average and RMS values are first-order approximations; the exact waveform is affected by the diode conduction threshold and source impedance.

TABLE II: Bridge rectifier values

Quantity	Value
Measured secondary RMS voltage	10.53 V
Secondary peak voltage	14.89 V
Assumed forward drop per diode	0.7 V
Bridge peak voltage	13.49 V
Rectified average voltage	8.59 V
Rectified RMS voltage	9.54 V
Input frequency	50 Hz
Full-wave ripple frequency	100 Hz
Ripple period	10 ms

Two commonly used rectifier diodes are the 1N4007 diode (1 A average forward-current class and 1000 V repetitive peak reverse-voltage rating) and the 1N5408 diode (3 A average forward-current class).

8. Smoothing Capacitor

The smoothing capacitor charges near the peaks of the rectified waveform and supplies the load between peaks. This reduces the voltage variation at the rectifier output.

The ripple calculation below uses a $2200\,\mu$F capacitor. For a full-wave rectifier with approximately constant load current and small ripple, the peak-to-peak ripple is approximated by

$$V_{r(pp)} \approx \frac{I_L}{f_r C}.$$

At $f_r = 100$ Hz and $C = 2200\,\mu$F, this gives

$$V_{r(pp)} \approx 4.545 I_L \text{ V},$$

where I_L is in amperes. Using the calculated peak of 13.49 V, the corresponding approximate average is

$$V_{\mathrm{DC}} \approx 13.49 - 2.273 I_L \text{ V}.$$

These expressions are analytical estimates and are not additional measurements.

For the separate centre-tapped transformer configuration, a $100\,\mu$F filter capacitor is specified, together with an approximately 18 V DC level for the 12 V RMS / approximately 18 V peak example. The $100\,\mu$F and $2200\,\mu$F values apply to different transformer and rectifier configurations and are therefore presented separately.

9. Regulator and Buck Converter

Two alternative regulation approaches are considered: the 7805 linear regulator and the LM2596 step-down switching regulator.

A. 7805 Linear Regulator

The 7805 is a fixed-output linear regulator with a nominal 5 V output. The TI LM340/LM7805 family documentation gives an approximately 2 V dropout figure at a stated 1 A test condition. The actual dropout depends on the specific device, load, and temperature.

The regulator's approximate power dissipation is

$$P_D \approx (V_{\mathrm{IN}} - V_{\mathrm{OUT}}) I_{\mathrm{OUT}}.$$

The 7805 family includes current limiting, thermal shutdown, and safe-operating-area protection. A current-limit figure of approximately 2.4 A describes protection behavior, not a continuous output-current rating. For this regulator family, capabilities above 1.5 A depend on the device variant and thermal conditions.

B. LM2596 Buck Converter

The LM2596 is a switching step-down regulator. Its family specifications include: input range 4.5–40 V, output-current capability up to 3 A under specified conditions, and a 150 kHz oscillator frequency. The current capability of a complete module also depends on its inductor, diode, capacitors, circuit board, and cooling.

For an ideal buck converter, the duty ratio is

$$D = \frac{V_{\mathrm{OUT}}}{V_{\mathrm{IN}}}.$$

A 12 V to 5 V example corresponds to an ideal duty ratio of approximately 41.7%. This is a reference calculation, not a measured operating point of the reported USB output.

TABLE III: Regulator comparison

Feature	7805	LM2596
Method	Linear	Switching buck
Output	Fixed 5 V	Fixed or adjustable
Main consideration	Headroom and heat	Ripple, layout, and heat
Key specifications	About 2 V dropout at stated 1 A test	4.5–40 V input; up to 3 A; 150 kHz

10. USB Output Port

The USB output includes a positive supply contact (VBUS), ground (GND), and interface contacts. In USB 2.0, D+ and D- carry data and may also participate in legacy charging-source detection. USB Type-C uses configuration-channel (CC) pins to advertise available current. USB Power Delivery (USB-PD) requires a compatible controller and negotiation; a fixed 5 V regulator alone does not provide USB-PD.

TABLE IV: USB charging current values

Port or mode	Current value
USB 2.0 SDP	Up to 500 mA*
USB 3.x SDP	Up to 900 mA*
BC1.2 CDP	Up to 1.5 A
BC1.2 DCP	Up to 1.5 A
USB Type-C current advertisement	1.5 A or 3 A at 5 V

**These are nominal maximum current values under the applicable USB specification and configuration/enumeration conditions; they are not guaranteed by the connector alone.*

The following table gives example legacy proprietary D+/D- voltage signatures. These voltage combinations are implementation-specific and are not universal USB charging rules.

TABLE V: Example legacy D+/D- charging signatures

D+	D-	Reported current
2.0 V	2.0 V	Approx. 0.5 A
2.0 V	2.7 V	Approx. 1.0 A
2.7 V	2.0 V	Approx. 2.1 A
2.7 V	2.7 V	Approx. 2.4 A

11. Measured USB Output

The USB output was measured using a voltmeter and an ammeter. The recorded average voltage was approximately 4.94 V, and the recorded average current was approximately 0.60 A. The measuring instrument also displayed 16 mAh. This value is recorded separately because mAh measures electric charge, whereas A measures current; the display value is not interpreted as an instantaneous current.

TABLE VI: Measured USB output values

Parameter	Measured value
Average output voltage	4.94 V
Average output current	0.60 A
Additional instrument display	16 mAh